

Not To Be Deceived? Timing Matters: Trustworthy Online Review Design

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ABSTRACT

Studies have shown that the effects of fake reviews can be decided by how platforms operate such as how they display online reviews. A meta-analytic study of communication research suggests that the timing of communication source identification affects message credibility. The current study suggests implementing reviewer information, used as criteria in a fake review detection algorithm, in online reviews and adjusting the identification timing of this information. The study findings show that 1) the number of accumulated helpful votes for the reviewer positively influences reviewer credibility, 2) perceived review authenticity mediates the relationship between reviewer credibility and users' intention to adopt the review, 3) reviewer identification timing affects the user's attitude about the product, such that viewing the reviewer's information alongside the review helps users consolidate their review evaluation. Implementing the recommended online review interface design based on the findings of this study can diminish the possible impact of fake reviews.

CCS CONCEPTS

• Human-centered computing; • Human-computer interaction (HCI); • Experimental studies in HCI;

KEYWORDS

Online review, Fake review, Review credibility, Trustworthy design

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1 INTRODUCTION

Before Black Friday, 2022, the Federal Trade Commission announced that it was exploring a potential regulation to combat deceptive review and endorsement practices, such as fake reviews [1]. The media also warned consumers of possible fake reviews [2] and announced several strategies to detect fake reviews [3]. Deceptive reviews and endorsements cheat consumers and lead them

to make uninformed purchase decisions. [1]. However, from consumers' perspective, it is difficult to differentiate fake reviews from genuine reviews. Even though mega review platforms, such as Yelp, have implemented fake review detection algorithms [7, 8], they do not guarantee perfect detection. In addition, many e-commerce sites and microsites do not control the dissemination of fake reviews. However, the effect of prevalent fake reviews can be controlled through the e-commerce review design, providing a more trustworthy online review interface. How fake reviews are displayed impacts how users interact with them. Given the impact of fake reviews, it is essential to give customers a more trustworthy online review interface by changing how online reviews are displayed [4]. To address this issue, we investigate two potential solutions for what information to show in the review and how to exhibit the review. Although important criteria for detecting fake reviews have been heavily investigated, online review platforms' efforts to make the factors more visible have not been adequately studied. It is also unclear when it is best to present review and reviewer information. This paper includes two studies. The first study aims to further understand important review characteristics that impact customers' judgment of reviewer credibility. The second study aims to identify feasible designs that enhance users' evaluation when engaging with reviews. The relationship between reviewer information and the identification timing is discussed to inform platforms about readers' intention to adopt the review and product attitude.

2 LITERATURE REVIEW

2.1 Fake online review and reviewer information

2.1.1 Fake online reviews. A fake online review is defined as review deceit from vendors, platform publishers, or any third-party monitoring online reviews and posting non-authentic online reviews to increase product sales [9]. As e-commerce grows, fake reviews have become more prevalent. Although estimates vary, between 10.3 percent [4] and 33.3 percent [10] of reviews are fake. Posting fake reviews is primarily motivated by financial gain or to gain an advantage over the competition [4]. Online merchants post positive fake reviews to promote their products or negative fake reviews to harm their competitors' business. However, fake reviews significantly affect stakeholders including consumers, merchants, and platforms. By increasing uncertainty and misleading consumers, fake reviews decrease consumers' trust in online reviews.

2.1.2 Fake review detection. Due to the prevalence of fake reviews, platforms have developed various fake review detection algorithms. These algorithms often utilize textual features in the review or are attuned to the behavioral features of reviewers. Recent research has identified the relative strength of reviewers' behavioral information

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in detecting fake reviews compared to the review texts [11–13]. Understanding reviewer behavior characteristics is important for two reasons. First, fake reviewers who are paid by third parties spend enough time polishing their reviews to make their reviews look more believable, and they make their reviews look similar to authentic reviews. Second, it is far more difficult for a reviewer to manipulate their review posting history or social interaction with others in the review platform than it is to manipulate the content [11]. Therefore, making reviewer information more salient in the online review will help users evaluate the credibility of the reviewer and the review.

2.1.3 Online reviewer information. Scholars have proposed various methods to analyze what reviewer information is most important when evaluating reviews and reviewer credibility. Studies have categorized reviewer information into four categories: personal, social, reviewing activity, and trusting. Reviewing activity is found to be the most helpful for users in detecting fake reviews [7]. This finding aligns with a study on hotel reviews that demonstrated how the accumulated useful vote count is the most important information in detecting fake reviews [11]. How long a reviewer has been active on an online platform is also an important reviewer behavioral signal that helps users detect fake reviews [12]. Another study describing user activity confirmed that if the reviewer has been active in a review platform longer, users' confidence in the authenticity of the reviewer increases [26]. Thus, we assume that the reviewer's tenure and accumulated helpful votes are important information for users to judge the credibility of the reviewer and the review.

H1. Longer reviewer tenure leads to higher perceived credibility of the reviewer than short reviewer tenure.

H2. A high number of accumulated helpful votes leads to a higher perceived credibility of the reviewer than a low number of accumulated helpful votes.

2.2 Reviewer identification timing

Online reviews provide diverse information about the product and the reviewer. Thus, platforms have developed separate reviewer pages. For example, the representative e-commerce Amazon and the popular review platform Yelp provide a separate reviewer page with more detailed information about the reviewer [23]. Thus, many customers are exposed to the review first, and then by clicking on the reviewer's name, users are directed to a page with more detailed reviewer information. This change provides different dynamics because the source and message, which are known to affect each other, are identified at different points in the user's journey. Although the effect of online review information has been studied in diverse fields, more research is needed regarding the design of the online review, specifically the exposure timing of the reviewer information. Traditional source credibility theory has demonstrated that people are more easily persuaded by messages written by credible sources than by non-credible ones. Source credibility is an important contributor to the persuasive effect of a message. However, the advantage of a high-credibility source over a low-credibility source is inconclusive [15, 16]. A meta-analytical review found that the timing of source identification influences the persuasive

effect of source credibility [17]. Low-credible sources are more persuasive when the source is revealed after the message is delivered rather than before the message is delivered [18]. Since the effect of source credibility tends to disappear when source information is introduced after message delivery, low credibility and no credibility sources (i.e., the source credibility is not mentioned) do not have a significantly different effect on the message persuasiveness or users' attitude change [19]. This finding implies that when online reviewer information is revealed after the review, people are more likely to trust the review, even if the reviewer's credibility seems questionable. One study showed that negative reviews compared to positive reviews reach larger audiences, and the effects of negative reviews persist longer than positive reviews [25], making the current online review design more vulnerable to users who are exposed to a negative fake review.

Therefore, to build a more trustworthy online review design, the timing of the reviewer information matters as well as the review and reviewer information. Considering the basic design of a typical websites and the placement of source information (i.e., either alongside or above the message), I hypothesize that when source information is presented alongside the review message, the persuasive effect of the message is potentially similar to the pre-message condition. Formally,

H3. Reviewer identification timing affects the intention to adopt the review; thus, pre- and same-message reviewer identification condition shows a more favorable intention to adopt the review than post-message reviewer identification condition, when the reviewer has a high number of accumulated helpful votes.

H4. Reviewer identification timing affects the intention to adopt the review; thus, pre- and same-message reviewer identification condition shows a less favorable intention to adopt the review than post-message reviewer identification condition, when the reviewer has low accumulated helpful votes.

2.3 Perceived authenticity

Authenticity refers to the sincerity, genuineness, and trustfulness of an object [20]. In the context of user experience, (e.g., a tourism study), the definition of authenticity encompasses the genuineness of the content [21]. For genuineness of the content in an online review, we define perceived authenticity as customers' subjective evaluation of the genuineness of online reviews. If the online reviewer is perceived as credible, the review will be perceived as authentic. Thus,

H5. Perceived review authenticity mediates the relationship between the perceived reviewer's credibility and the intention to adopt the review.

3 METHOD AND RESULT

3.1 Study 1. Which reviewer information to show in the online review

3.1.1 Design, Participants, and procedure. To test hypotheses 1 and 2, the study employed a 2 (Reviewer tenure: long vs. short) x 2 (Accumulated helpful votes: high vs. low) between-subjects

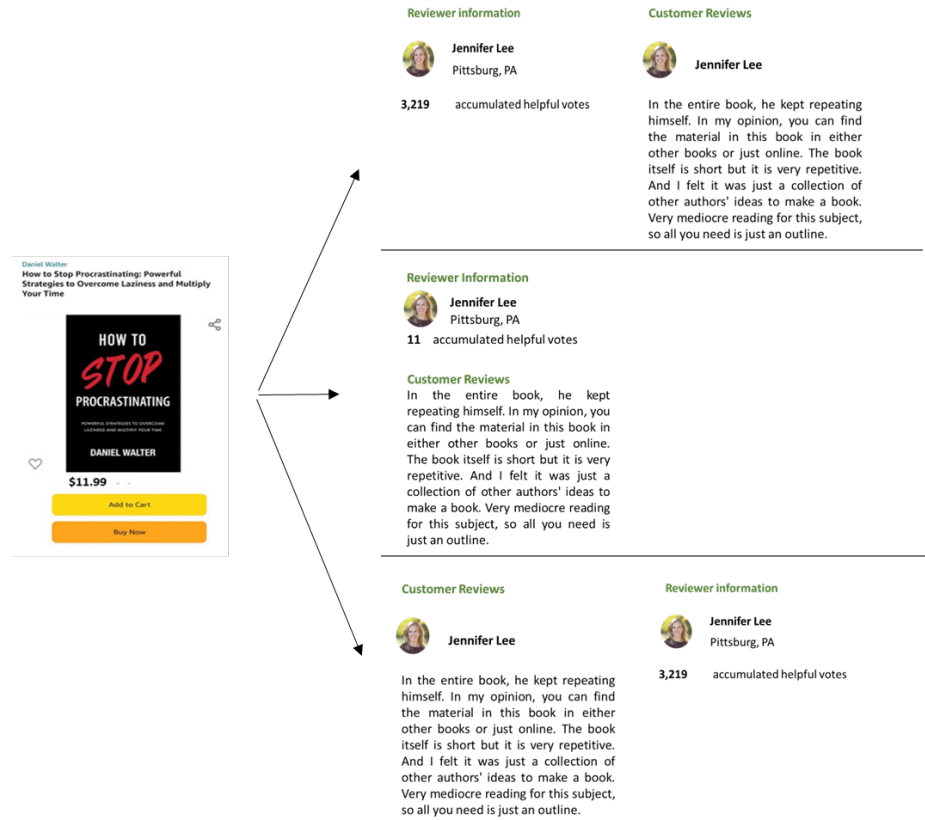


Figure 1: Product information and pre, same, and post-message reviewer identification condition

design. Participants ($n=265$) were recruited through CloudResearch and provided with monetary compensation. The participants were instructed to imagine that they were using an online commerce site and were exposed to the product page and online reviews in e-commerce. The product is a non-fiction book named 'Stop Procrastinating' priced at \$11. Online review content was the same across all the conditions, and reviewer tenures and accumulated helpful votes were manipulated. The participants, exposed to one of four conditions, answered the manipulation check questionnaire. After the exploration of provided online review, participants answered questions asking about their perception of the reviewer.

3.1.2 Manipulation check. First, we measured whether the participants remembered seeing the reviewer's information. After that, they were asked whether they think the reviewer has been in this review platform long or not and whether they think the number of accumulated helpful votes the reviewer got is many or not.

3.1.3 Result. Manipulation check result for reviewer tenure showed that participants in the high-reviewer tenure condition perceived the reviewer tenure ($M = 5.3$, $SD = 1.4$) as longer than participants in the low-reviewer tenure condition ($M = 3.6$, $SD = 1.9$) $F(1, 263) = 65.9$, $p < .001$. A check of accumulated helpful votes manipulation showed that participants in the high-helpful votes condition rated the number of accumulated helpful votes ($M = 5.7$, $SD = 1.2$) higher than participants in the low-helpful vote condition

($M = 3.8$, $SD = 1.6$) $F(1, 263) = 115.1$, $p < .001$. To test hypotheses 1 and 2, univariate analysis was used, controlling non-fiction book involvement. The analysis revealed that participants with high accumulated helpful votes rated significantly higher perceived reviewer credibility ($M = 5.4$, $SD = 1.0$) than participants with low accumulated helpful votes ($M = 5.0$, $SD = 1.0$) $F(1, 262) = 9.2$, $p < .05$, partial $\eta^2 = 0.03$. Thus, H1 is supported. However, participants in high tenure didn't rate significantly higher perceived reviewer credibility ($M = 5.3$, $SD = 1.0$) than participants in low tenure condition ($M = 5.2$, $SD = 1.0$) $F(1, 262) = 0.3$, $p > .05$, partial $\eta^2 = 0.001$. Thus, H2 is not supported. Therefore, we decided to add only accumulated helpful votes as an independent variable in study 2.

3.2 Study 2. How to exhibit the reviewer's information

3.2.1 Design, Participants, and procedure. To test hypotheses 3, 4, 5, the study employed a 2 (Accumulated helpful votes: high vs. Low) $\times$ 3 (Reviewer information identification timing: Pre vs. Post vs. Same message) between-subjects design. A total of 254 students were recruited from CloudResearch in the experiment. After answering the consent form, participants were assigned to the same product information page as study 1. Considering the influence of negative reviews over positive reviews, the negative review was selected as a stimulus. After seeing the product information, the

participants were randomly assigned to one of six conditions. Reviewer information has the number of accumulated helpful votes information with the reviewer's name and where they are from. Pre- and post-message source identification condition stimuli are composed of two separate pages of the app, so participants need to go to the next page by clicking the next button. Therefore, the participants were exposed to the reviewer information and review information on a separate page. The same message source identification condition has review and reviewer information on the same page. After seeing the e-commerce product page with the product review, they answered questions asking about their credibility perception toward the reviewer, the authenticity of the review, and the persuasiveness of the message such as intention to adopt the review and product attitude. Lastly, they were provided with the code for claiming compensation for their participation.

3.2.2 Manipulation check. First, we measured whether the participants remembered seeing the reviewer's information. After that, they were asked whether they think the number of accumulated helpful votes the reviewer got is many or not. Lastly, for the manipulation check of source identification timing, the participants were asked whether they had seen the reviewer's information before or after the review, or were on the same page with reviews.

3.2.3 Result. Data from 181 participants who passed the manipulation check were used for hypothesis testing. H3 hypothesized that the pre-and same-message reviewer identification condition shows a more favorable intention to adopt the review than the post-message reviewer identification condition when the reviewer has a high number of accumulated helpful votes. Even though the pattern of intention to adopt the review was the same as hypothesized such as intention to adopt the review in pre-message ($M = 5.4$, $SD = 1.3$) is highest, followed by the same message ($M = 5.1$, $SD = 1.2$), and post-message ($M = 4.9$, $SD = 1.3$), it was not statistically significant. Thus, H3 is not supported. H4 hypothesized that the pre-and same-message reviewer identification condition shows a less favorable intention to adopt the review than the post-message reviewer identification condition when the reviewer has low accumulated helpful votes. Intention to adopt the review showed the highest value in post-message ($M = 5.1$, $SD = 1.3$) followed by pre-message ($M = 5.0$, $SD = 0.9$), and the same message ($M = 4.7$, $SD = 1.1$), but it was not statistically significant (Figure 2). Thus, H4 is not supported. H5 hypothesized that the perceived review authenticity mediates the relationship between the perceived reviewer's credibility and the intention to adopt the review. To test hypothesis 5 PROCESS macro version 4.1 was used, controlling for issue involvement. The result revealed the indirect effect of perceived reviewer credibility on intention to adopt the review through perceived review authenticity was significant ($B = .46$, $SE = .09$, 95% confidence interval [C.I.] from .28 to .64). Therefore, H5 is supported.

To understand whether users' intention to adopt the review is reflected in their product attitude, we also measured product attitude. Interestingly, in the low accumulate helpful vote condition, participants in the pre-message condition rated the least favorable product attitude ($M = 2.8$, $SD = 1.2$), followed by pos-message ($M = 3.6$, $SD = 1.5$) and same-message ($M = 3.9$, $SD = 1.6$) $F(2, 89) = 3.8$, $p < .05$, partial $\eta^2 = 0.79$ (Figure 3). Therefore, even though reviewer identification timing didn't have a significant effect on

users' intention to adopt the review, but it had a significant effect on product attitude.

4 DISCUSSION

The study presents interesting findings. First, the reviewer information used to detect fake reviews influences the perceived credibility of the reviewer. As fake review algorithm literature has shown, accumulated helpful votes of the reviewer significantly affects participants' perceived reviewer credibility. However, in this study, reviewer tenure, found to be an important signal to detect fake reviews in the fake review algorithm, did not significantly affect perceived reviewer credibility. This finding shows that customers' understanding of online review credibility does not necessarily reflect what the algorithm found. Second, the reviewer identification timing had a statistically significant impact on users' attitude about the product when the reviewer is less credible. When the low credible reviewer was identified prior to the review, participants' attitude about the product was significantly lower than when the low credible reviewer was identified after the review. Considering that the participants were exposed to negative reviews, the result indicates that the low credibility of the reviewer was not reflected in the participants' message evaluation. This finding is different from the meta-analytic findings of previous literature, arguing that the effect of the source credibility is more salient when the source information is identified prior to the message. Considering the offline communication experiment conditions of the meta-analytic findings, our study result shows that users' perceptions of the reviewer's credibility did not persist if the review was presented separately in the online interface. Thus, platform designers need to consider how to present relevant reviewer information to users not only at the order level on the same page but also at the page level. Third, the perceived review authenticity elicited by perceived reviewer credibility positively affected users' intention to adopt the review indicating that displaying meaningful reviewer information will help users make better reviewer credibility judgments. In addition, this evaluation will be a criterion for review authenticity, ultimately influencing review persuasiveness. Lastly, the impact of reviewer identification timing is more salient when the reviewer is less credible than more credible. Considering that the impact of negative reviews is more influential than positive reviews, identifying less credible reviewers alongside the review will help users make informed decisions about products. Since online reviewers have started to provide more information on platforms, platform designers must determine what information needs to be presented and when to show it in the user journey. Instead of hiding all the reviewer information on a separate reviewer page that users can only access after reading the review, displaying reviewers' accumulated endorsement information alongside the review would reduce the impact of fraudulent reviews from less credible reviewers. These findings shed light on a more trustworthy online review interface design so users are less vulnerable to fake reviews.

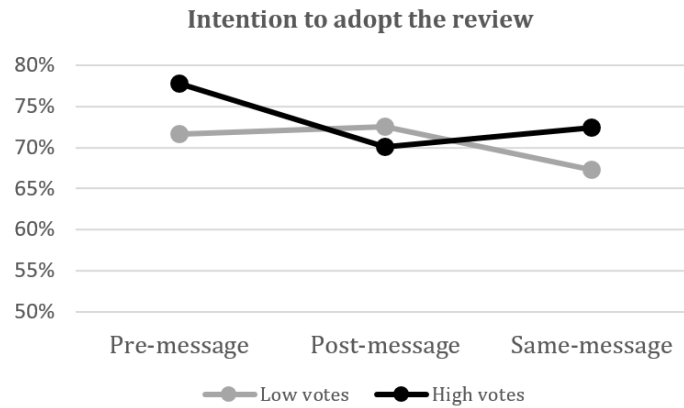


Figure 2: Intention to adopt the review as a function of different reviewer identification timing $F(2, 174) = 1.1$, $p = .34$, $\eta^2 = 0.1$.

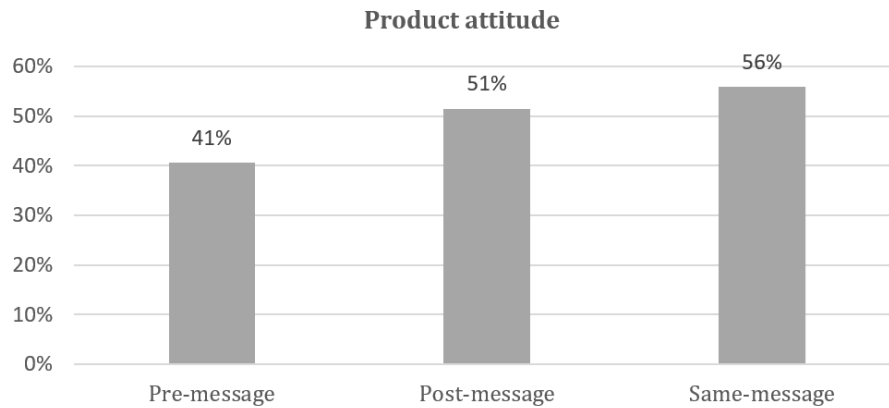


Figure 3: Product attitude as a function of different reviewer identification timing when the reviewer got a low number of helpful votes. This figure shows that reviewer identification timing significantly affects product attitude when the reviewer is not credible. $F(2, 89) = 3.8$, $p < .05$, partial $\eta^2 = 0.79$

5 LIMITATIONS AND FUTURE WORK

The study provides interesting insights reflecting meta-analytic knowledge from communication literature on e-commerce interfaces. However, it is not free from limitations. First, in the experimental condition, participants were only shown a single review. To verify the external validity of the results, future studies should examine participants' responses in a real e-commerce interface setting with more reviews. Second, we selected a non-fiction book as the stimuli because online reviews have been found to be more influential for experience products than search products. Although we controlled for users' involvement in non-fiction books, the book context may have affected participants' product attitudes. Follow-up studies can verify the results in other product categories. Lastly, we only tested the effect of reviewer identification timing in the negative review condition. The results may be easier to generalize if the effect is tested in a positive review. Future empirical studies could also test the effect of reviewer identification timing in different settings in terms of review valence, and address how the e-commerce interface can help users make better credibility judgments about reviewers and reviews.

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Table 1: Intercorrelations Among Measured Variables

	1	2	3	4	5
1. Issue Involvement	-				
2. Perceived Reviewer Credibility	.08	-			
3. Perceived Review Authenticity	.14	.75**	-		
4. Product Attitude	.09	-.10	-.24	-	
5. Intention to Adopt the Review	.03	.65**	.71**	-.26**	-

** Correlation is significant at the 0.01 level (2-tailed).

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A APPENDICES

In the appendix section, three levels of Appendix headings are available.

Table 2: Measurement items

Construct	Measurement Items	M	SD	A
Issue Involvement	To me, a non-fiction book is Unimportant – important Means nothing to me – means a lot to me Not needed – needed Doesn't matter to me – matters to me	5.6	1.3	.92
Perceived reviewer credibility	This reviewer is trustworthy. This reviewer is reliable. This reviewer is reputable. This reviewer seems like an expert reviewer. The reviewer is qualified. The reviewer is informed. The reviewer is competent.	5.2	1.0	.95
Review authenticity	This online review provides me with authentic information. This online review provides me with natural information. This online review provides me with genuine information. This online review provides me with true information	5.4	1.1	.93
Product attitude	The book 'How to stop procrastinating' seems likable I feel positive about the book 'How to stop procrastinating' personalized advertisements. The book 'How to stop procrastinating' seems credible	3.5	1.5	.94
Intention to adopt the review	The information in the online review motivates me not to purchase the book 'How to stop procrastinating'. I would like to follow the suggestions in online reviews. I would like to adopt this online review.	5.0	1.2	.82